Customer Behavior & Purchase Trend Analysis

1. Project Overview

This project looks at 3,900 customer shopping records to understand how people buy, what they prefer, and how much they spend. By exploring the data and creating clear visuals, we find useful patterns and important customer groups. These insights help improve marketing, target the right customers, and support better business decisions.

2. Data Summary

The dataset contains **3,900 customer records** and **18 columns**. The **Review Rating** column has **37 missing values**.

Key features:

Customer Demographics:

These features describe who the customer is and help identify segments based on age, gender, and location. Understanding demographics allows businesses to target the right audience and tailor marketing strategies. (Customer ID, Age, Gender, Location)

Purchase Details:

These features explain what customers buy, how much they spend, and the characteristics of their purchases. They help analyze product demand, spending patterns, seasonal trends, and satisfaction through review ratings. (Item Purchased, Category, Purchase Amount (USD), Size, Color, Season, Review Rating, Shipping Type, Payment Method)

Shopping Behavior:

These features reflect how customers shop, their loyalty, and their purchasing frequency. They help identify repeat buyers, discount usage trends, and overall shopping habits, which support customer retention strategies. (Subscription Status, Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases)

3. Exploratory

Data Analysis using Python We began with data preparation and cleaning in Python:

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used df.info() to check structure and .describe() for summary statistics.

```
df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900.000000	3900	3900
2	NaN	6	7
No	NaN	PayPal	Every 3 Months
2223	NaN	677	584
NaN	25.351538	NaN	NaN
NaN	14.447125	NaN	NaN
NaN	1.000000	NaN	NaN
NaN	13.000000	NaN	NaN
NaN	25.000000	NaN	NaN
NaN	38.000000	NaN	NaN
NaN	50.000000	NaN	NaN

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

● Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

● Column Standardization: Renamed columns to snake case for better readability and documentation.

● Feature Engineering:

- ○ Created age_group column by binning customer ages.
- ○ Created purchase_frequency_days column from purchase data.

● Data Consistency Check: Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

● Database Integration: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender text	total_revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	96	100
2	616	100
3	582	100
4	1592	100
5	194	100
6	519	100
7	862	100
8	770	100
9	244	100
10	1480	100
Total rows: 839	Query complete 00:00:00.108	

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased text	rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	purchase_amount numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status text	subscription_count bigint	avg_purchase_amount numeric	total_revenue numeric
1	No	2847	59.87	170436
2	Yes	1053	59.49	62645

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_apply numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of customer bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	total_customer bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

11. **Customers Purchasing in Winter** – Retrieved all customers who made purchases during the Winter season.

	total_customer_in_winter bigint
1	971

12. **Average Customer Age** – Calculated the mean age of all customers in the dataset.

	Average age(years) numeric
1	44.07

13. **Customers by Location** – Filtered and identified customers from a selected location such as "New York."

	customer_id bigint
1	15
2	81
3	99
4	106
5	111
6	196
7	223

Total rows: 87 Qu

14. **Revenue by Product Category** – Computed total revenue contribution for each product category.

	category text	revenue numeric
1	Clothing	104264
2	Accessories	74200
3	Footwear	36093
4	Outerwear	18524

15. **High Review Ratings** – Retrieved customers who gave a review rating above 4.

	customer_id bigint	review_rating double precision
1	10	4.8
2	11	4.1
3	12	4.9
4	13	4.5
5	14	4.7
6	15	4.7
7	17	4.1
8	18	4.7
Total rows: 1453	Query complete 00:	

16. **Express Shipping Usage** – Calculated the percentage of purchases made using the "Express" shipping type.

	Express Shipping Usage in % numeric
1	16.56

5. Dashboard in Power BI

I built an interactive dashboard in Power BI to present insights visually

6. Business Recommendations

1. Boost Subscription Conversion

With only **27% of customers subscribed**, there is strong potential to grow recurring revenue. Offer exclusive perks such as early access, member only discounts, or free express shipping to increase signups.

2. Strengthen Customer Loyalty Programs

The data shows a steady revenue contribution from repeat customers. Launch a structured loyalty system reward points, tier levels, or cashback to motivate more customers to shift into the loyal and high frequency shopping segment.

3. Review and Optimize Discount Policies

A significant portion of purchases involve discounts, but margins must be protected. Analyze discount effectiveness by product category and limit unnecessary price cuts. Focus on targeted promotions for high value categories like **Clothing** and **Accessories**.

4. Leverage Product Positioning & Best Sellers

Products like **Blouses, Shirts, Dresses, Pants, and Jewelry** contribute over $10K each. Highlight these top performing items in marketing campaigns to maximize sales. Pair them with frequently bought together recommendations to increase average order value.

5. Target High-Revenue Customer Segments

The **Young Adult** and **Middle aged** groups generate the highest revenue. Tailor personalized marketing campaigns, seasonal offers, and product suggestions to these age groups to drive more conversions.

6. Improve Shipping Strategy

Users choosing faster shipping options (like Express) show stronger purchase intent. Offer shipping-based incentives such as free express shipping for orders above a threshold to increase both order value and customer satisfaction.

7. Expand High-Performing Categories

Clothing and Accessories dominate revenue, contributing more than 75% combined. Expand these categories with more variety and sizes. Optimize stock based on demand patterns to reduce lost sales and inventory gaps.

8. Enhance Review & Feedback Engagement

With an **average rating of 3.75**, there is room for improvement. Encourage more customer reviews with small incentives and use the feedback to refine product quality and service.